



THE UNIVERSITY OF  
MELBOURNE

# Marketing Research

Week 12

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# Housekeeping

- The rubric is available on Canvas (under Group Project)
- Note that in the section of quantitative analysis, you do not need to follow the order of the tests (i.e., H1 = Chi-Square, ...). Instead, you can order the test based on the logical flow of your report.
- Chooses and applies appropriate techniques (e.g., figures, summary statistics, tables, hypothesis tests) with some justification. Ensures the assumptions of the methods are met. If you feel it necessary, explain why the method was appropriate given the data structure and measurement levels.



# Common questions

- Q: Do I need to use all of the survey questions for the report?
- A: You do NOT need to use all of the survey questions for the report. Select the important ones in the written report or leave them in the appendix.
- Q: Is it okay to report insignificant results?
- A: It is OKAY to report insignificant results that do not support your hypotheses (e.g., you do NOT detect any difference in satisfaction between two gender groups). However, you should provide reasoning why it is still interesting to report null results. (see next page for further explanation)



# Hypothesis testing: Dealing with non-significance

Seeing lots of non-significant (NS) results? Don't panic, it's just a normal part of research. Continue exploring the data and develop new hypotheses to test if needed.

Do include a mix of significant and non-significant findings in your report.

- You will still get the full Results interpretation marks as long as you interpret your NS output correctly.
- You do get extra Discussion/Critical analysis points for explaining potential reasons for your NS findings.



## Example: Writing up non-significant results

“We conducted an independent-samples t-test to test H2. There is a no significant difference in purchase intention between physically active and non-active customers,  $t(38) = -1.397, p > .05$ . This can be further confirmed by checking the mean figures for the two sample groups,  $M_{active} = 4.39, M_{non-active} = 4.41$ . These results do not support H2. We therefore do not reject the null hypothesis.”

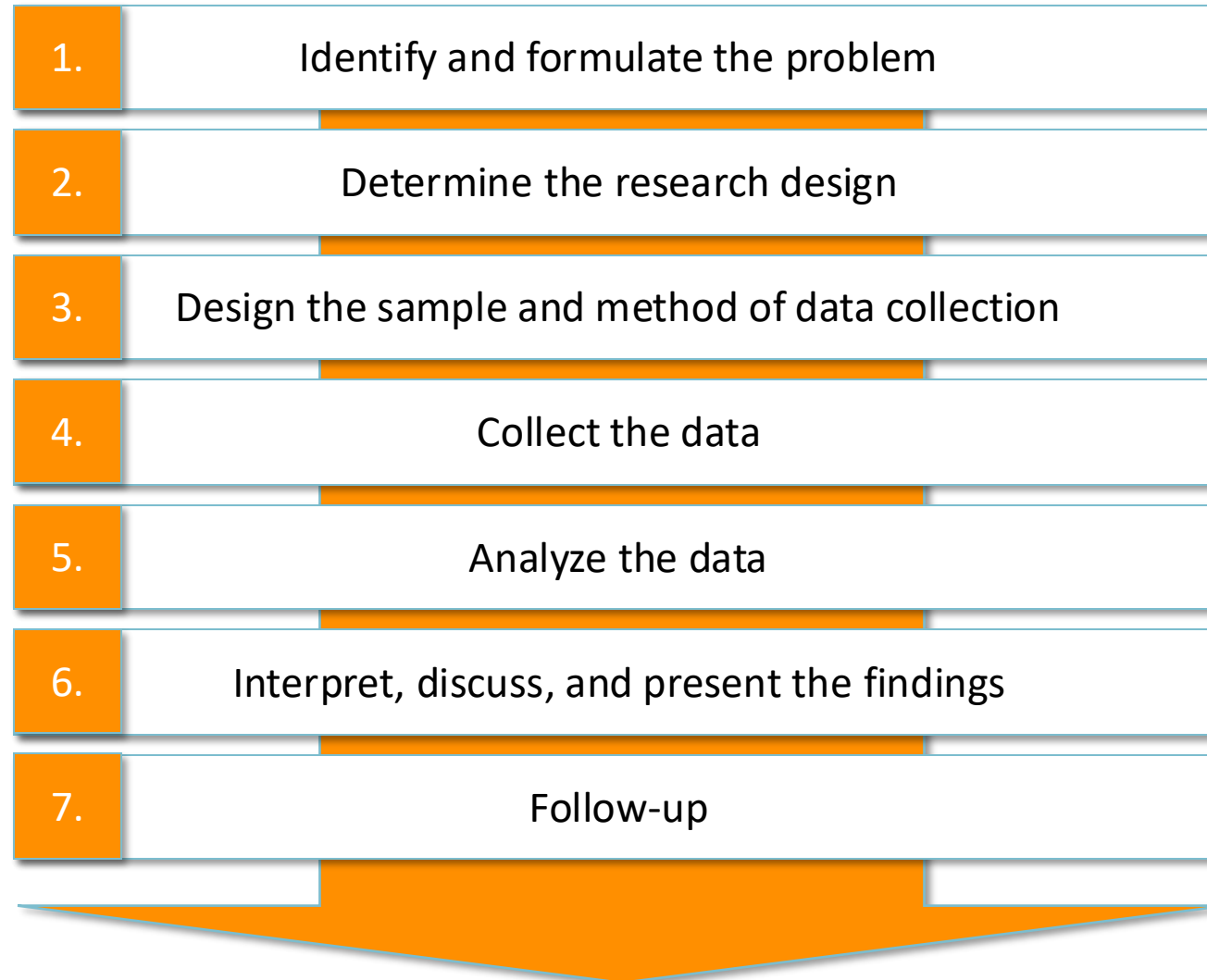


## Common questions

- Q: Why is my p-value in ANOVA smaller than 0.05, but I don't see any pair comparisons with a p-value smaller than 0.05?
- A: In ANOVA, this is usually caused by the small sample size. Refer to the next question (e.g., borderline p-values) for potential solutions.
- Q: What to do with borderline significant p-value (e.g.,  $p = 0.071$ ).
- A: By default, we assume a significance level of  $\alpha = 0.05$  for hypothesis tests. However, due to small sample sizes, you might find that many p-values in your SPSS outputs are borderline (e.g.,  $p = 0.071$ ). There are two ways to deal with them:
  - Interpret the results assuming  $\alpha = 0.05$ ; you can indicate that you do not reject the null hypothesis at the 95% significance level; however, it is significant at the 90% significance level.
  - You can **explicitly** set  $\alpha$  to be a slightly larger value (e.g.,  $\alpha = 0.15$ ). In this case, you can say that you reject the null hypothesis at the **85%** significance level.



# The Market Research Process





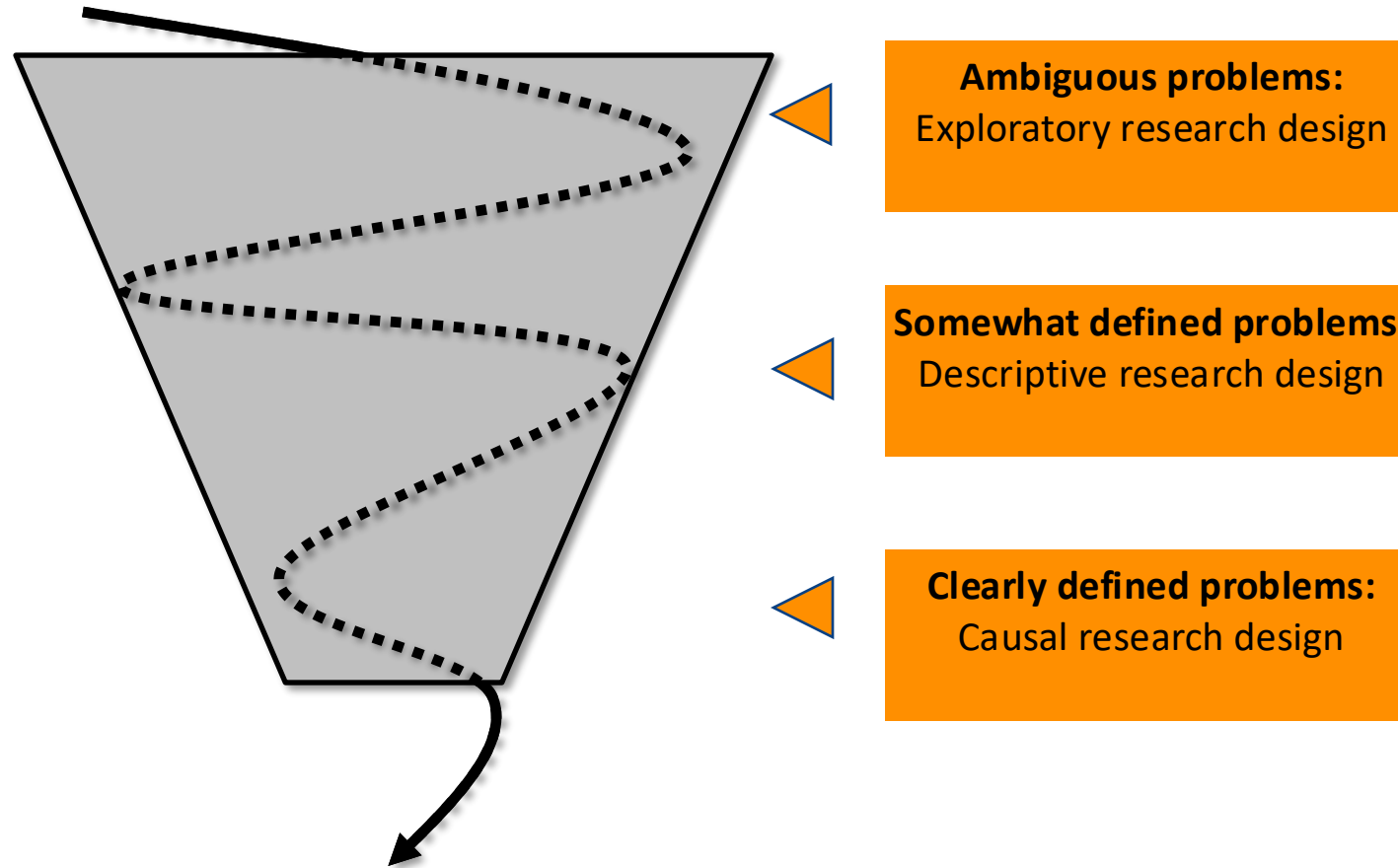
## Question:

Why do marketing researchers use exploratory research?

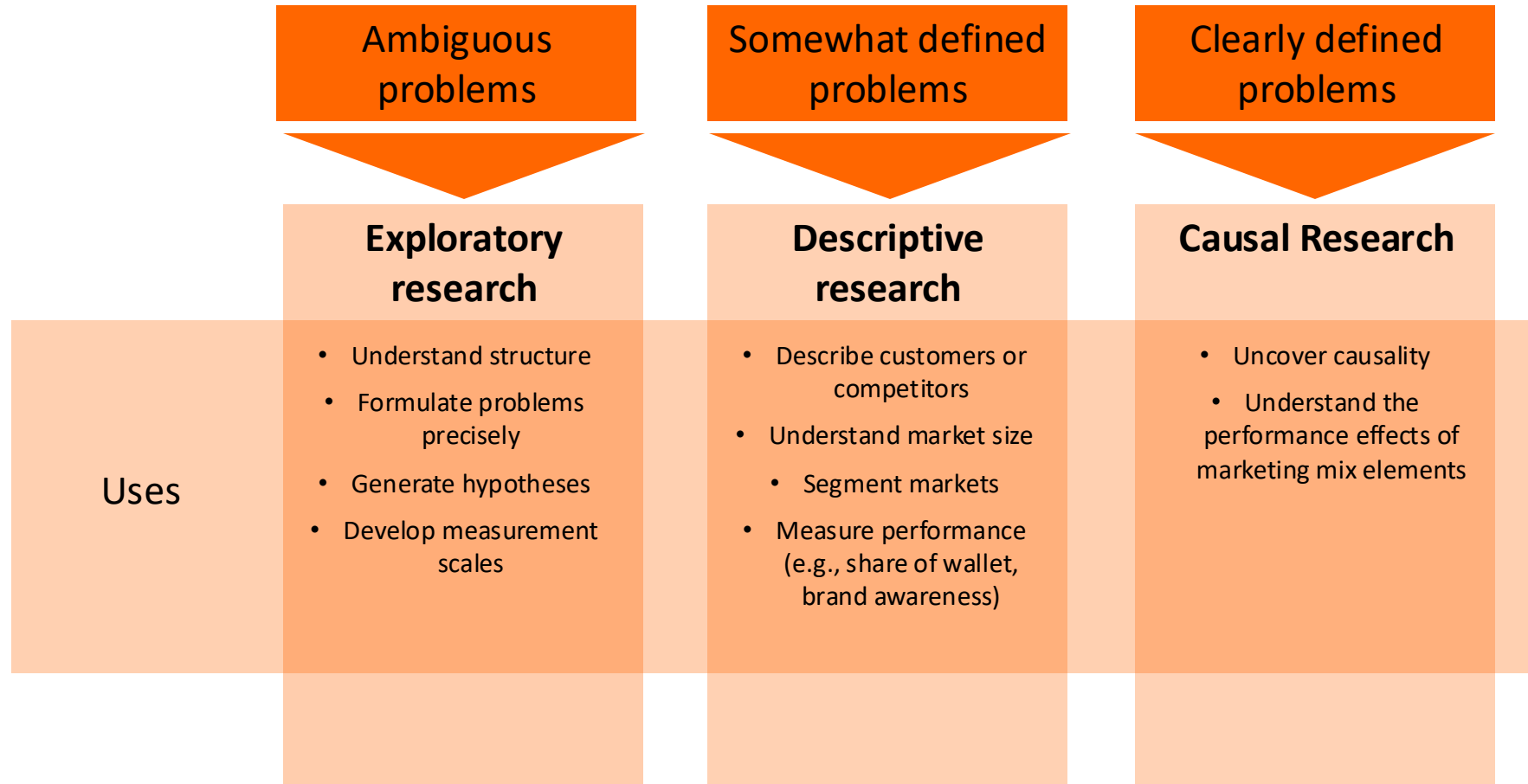
- a. To clarify and gain an understanding of a market segment
- b. To test hypotheses
- c. To measure market share
- d. To predict future sales



# Second Step: Determine the Research Design



# Determine the Research Design





## Question

Which of the following is NOT a purpose of focus groups?

- a. Develop creative concepts for a promotional campaign
- b. Obtain impressions of a new product
- c. Conduct a statistical analysis of consumer behavior
- d. Generate new ideas for a product design



## Question

During a series of in-depth interviews, customers of a fashion rental platform repeatedly describe the service using terms like “freedom,” “no commitment,” and “variety.” What should the researcher do next in the qualitative analysis?

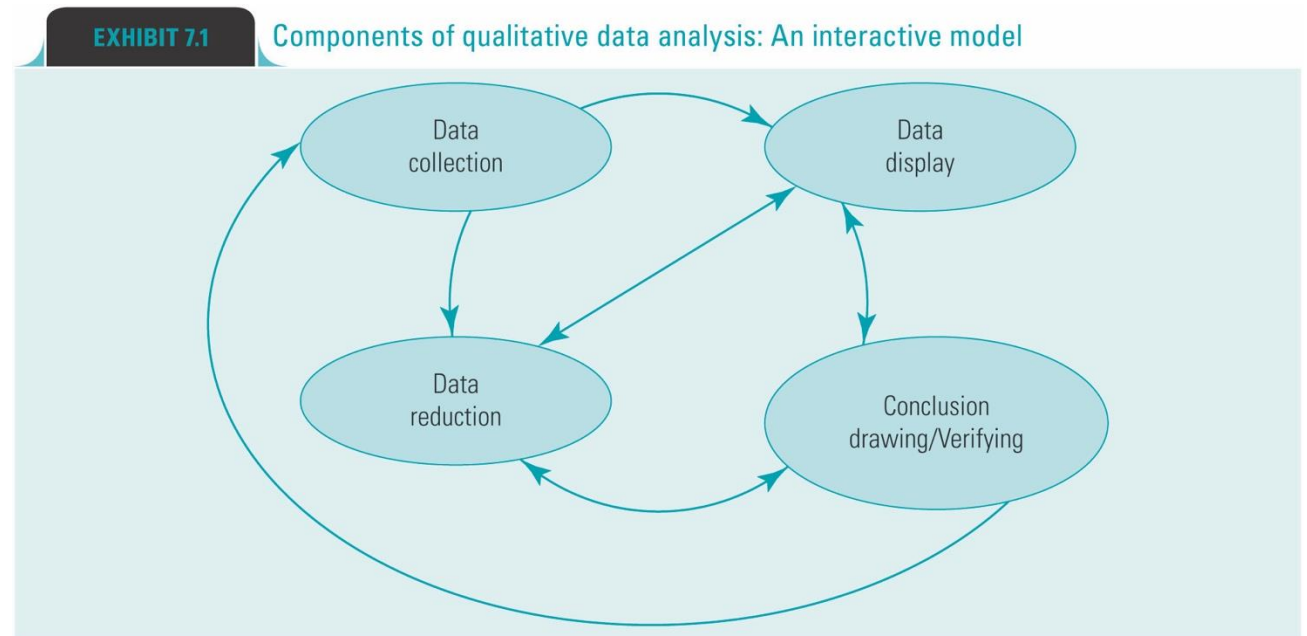
- a. Calculate the mode of the most used term
- b. Code and cluster these terms into emerging themes
- c. Discard repeated words as redundant
- d. Run a one-way ANOVA to confirm associations

# Exploratory research and qualitative analysis

## Choices of exploratory research methods

- Focus groups
- Depth interviews
- Projective techniques
- Observational studies
- Ethnographies

## Qualitative Data Analysis



Source: Matthew B. Miles & A. Michael Huberman, *Qualitative Data Analysis: An Expanded Sourcebook*, Sage Publications, Thousand Oaks, CA, 1994, p. 12.



# Types of Data

What characterizes data?

- **Variable:** an attribute whose value can change. A *variable* is the opposite of a *constant*.
- A **case** (or observation) consists of all the observed variables that belong to an object such as a customer, a company or a country.
- **Item:** usually refers to a survey question put to a respondent

## Constructs

- Construct: a variable that is not directly observable. Examples:
  - Happiness
  - Trust
  - Satisfaction
- Not all constructs (or construct measures) are alike:
  - Reflective constructs versus formative constructs
  - Single- versus multi-item constructs
- Developing constructs requires using scale/index development procedures



# Question

What is the main difference between primary and secondary data?

- a. Primary data is always more reliable than secondary data
- b. Primary data is collected for the specific research purpose; secondary is previously collected for another purpose
- c. Secondary data is always more recent than primary data
- d. Primary data is always quantitative, secondary is always qualitative



# Advantages and Disadvantages of Secondary and Primary Data

|               | Secondary data                                                                                                                                                                                                                                                                                          | Primary data                                                                                                                  |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Advantages    | <ul style="list-style-type: none"><li>• Tends to be cheaper</li><li>• Sample sizes tend to be greater</li><li>• Tend to have more authority</li><li>• Are usually quick to access</li><li>• Are easier to compare to other research using the same data</li><li>• Are sometimes more accurate</li></ul> | <ul style="list-style-type: none"><li>• Are recent</li><li>• Are specific for the purpose</li><li>• Are proprietary</li></ul> |
| Disadvantages | <ul style="list-style-type: none"><li>• May be outdated</li><li>• May not fully fit the problem</li><li>• There may be hidden errors in the data</li><li>• Usually contain only factual data</li><li>• No control over data collection</li><li>• May not be reported in the required form</li></ul>     | <ul style="list-style-type: none"><li>• Are usually more expensive</li><li>• Take longer to collect</li></ul>                 |

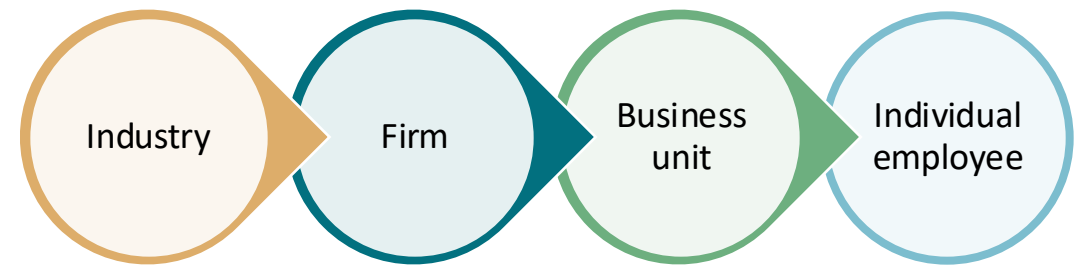




# Unit of Analysis

Unit of analysis is the level at which a variable is measured

- Customer satisfaction
- Profitability



Data is best collected at the lowest level

- Data can be aggregated (summed) to higher levels later
  - Example: “Cars sold at each individual car dealership” can be aggregated to “Cars sold in region / country XYZ”



# Dependent and Independent Variables

Dependent and independent variables are terms relating to the design of a market research study

**Dependent variable**

- What you try to explain

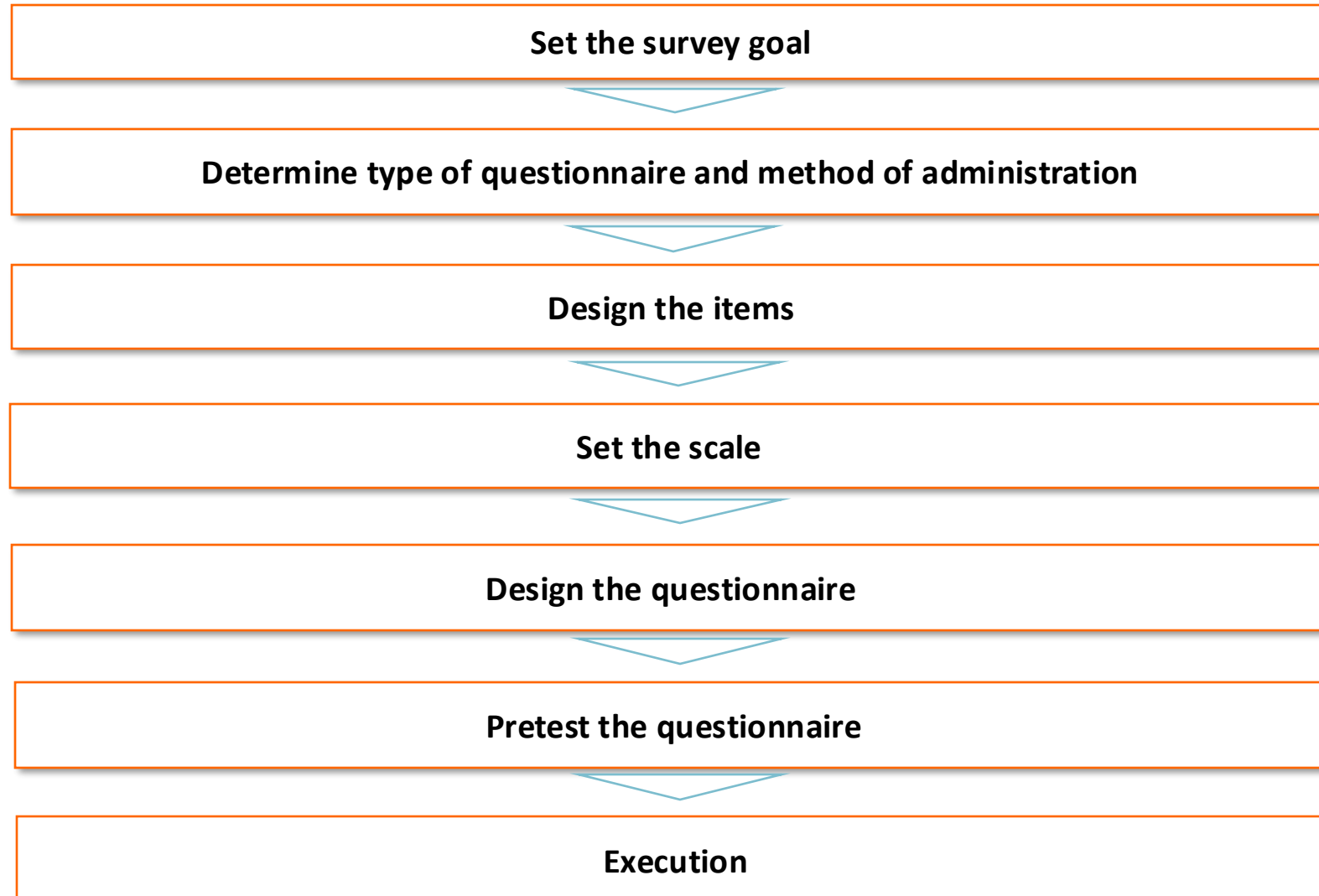
**Independent variable**

- What you use to explain it

Do not confuse with 'does the variable depend on something else.' Almost all variables depend on something else (e.g., loyalty, price sensitivity etc.)



# Collecting Quantitative Data: Designing Questionnaires





# Question

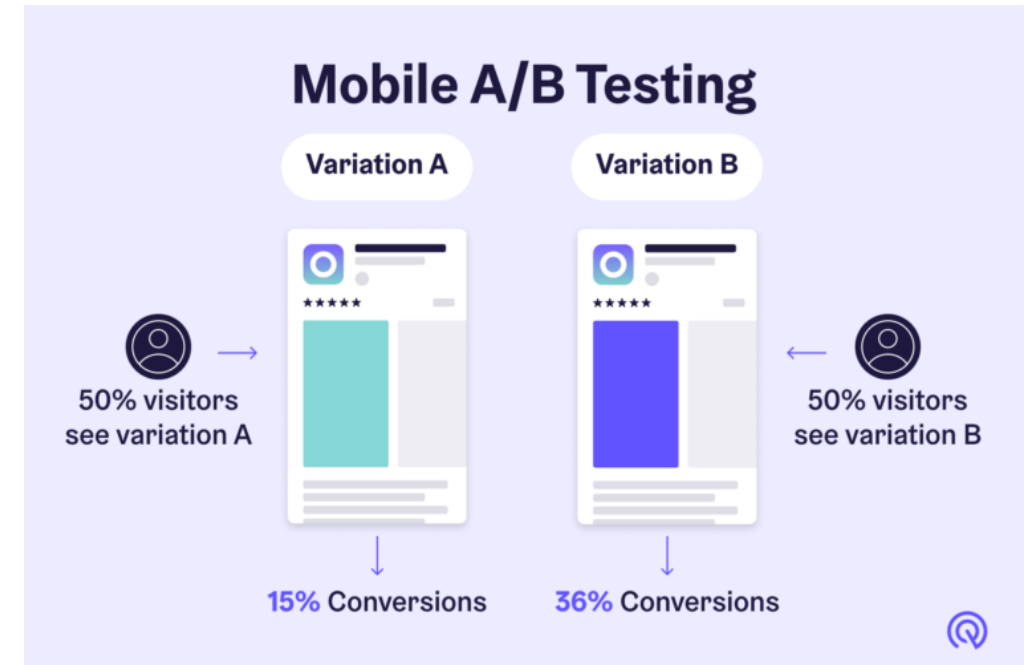
Which type of experiment is more likely to have high external validity?

- a. Lab experiments
- b. One-shot case studies
- c. Field experiments
- d. Before-after designs without control groups



# Causal Research

- Correlation vs causality
- Experiments help establish causality
  - Lab experiments
  - Field experiments
  - Test markets
- Internal vs external validity





# Experimental Designs

|                             |   |                |
|-----------------------------|---|----------------|
| <b>One-shot case study:</b> | X | O <sub>1</sub> |
|-----------------------------|---|----------------|

|                             |                |   |                |
|-----------------------------|----------------|---|----------------|
| <b>Before-after design:</b> | O <sub>1</sub> | X | O <sub>2</sub> |
|-----------------------------|----------------|---|----------------|

|                                                          |                        |                |   |                |
|----------------------------------------------------------|------------------------|----------------|---|----------------|
| <b>Before-after experiment<br/>with a control group:</b> | Experimental group (R) | O <sub>1</sub> | X | O <sub>2</sub> |
|                                                          | Control group (R)      | O <sub>3</sub> |   | O <sub>4</sub> |

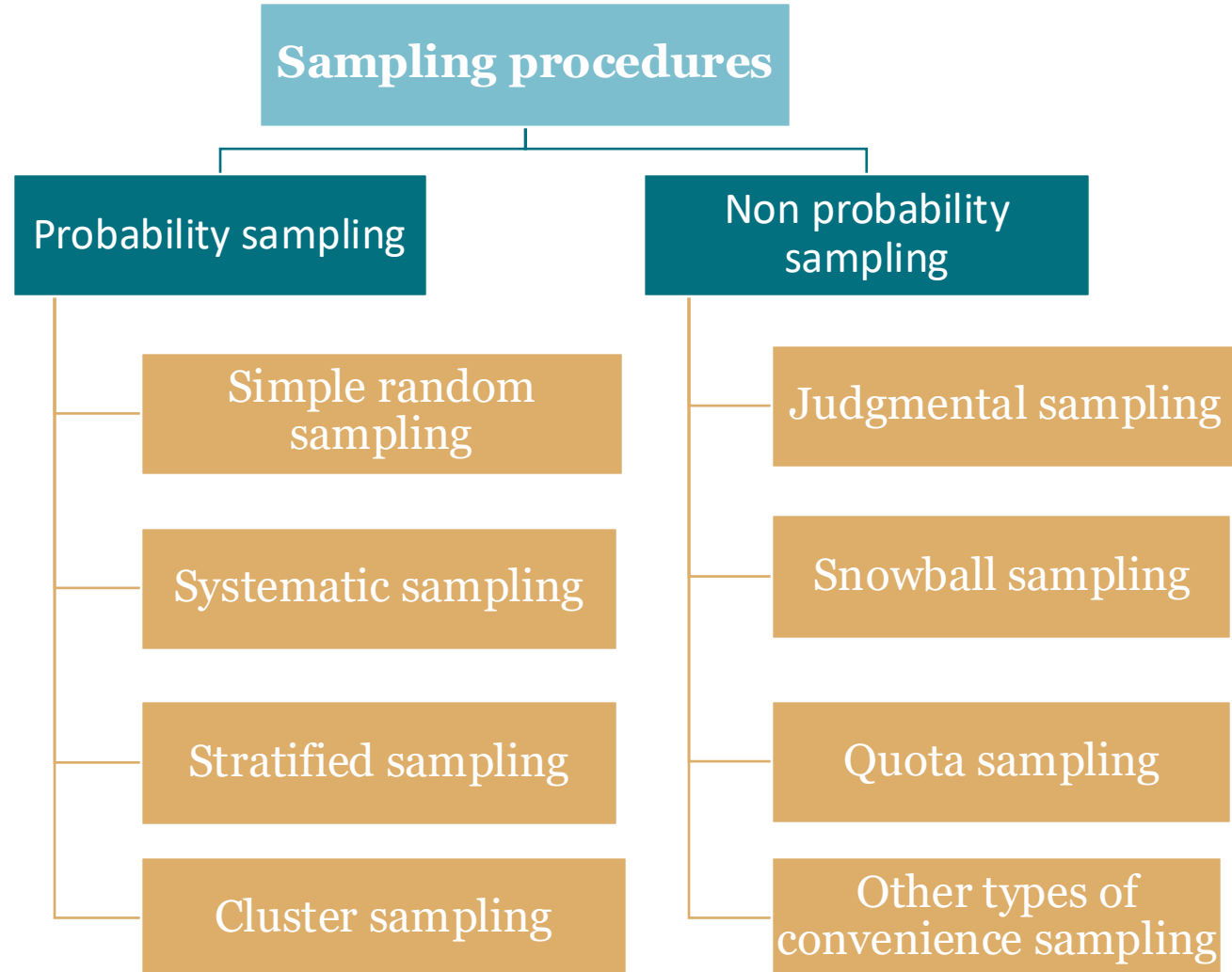


## Question

In which type of probability sampling is every  $k$ -th element selected?

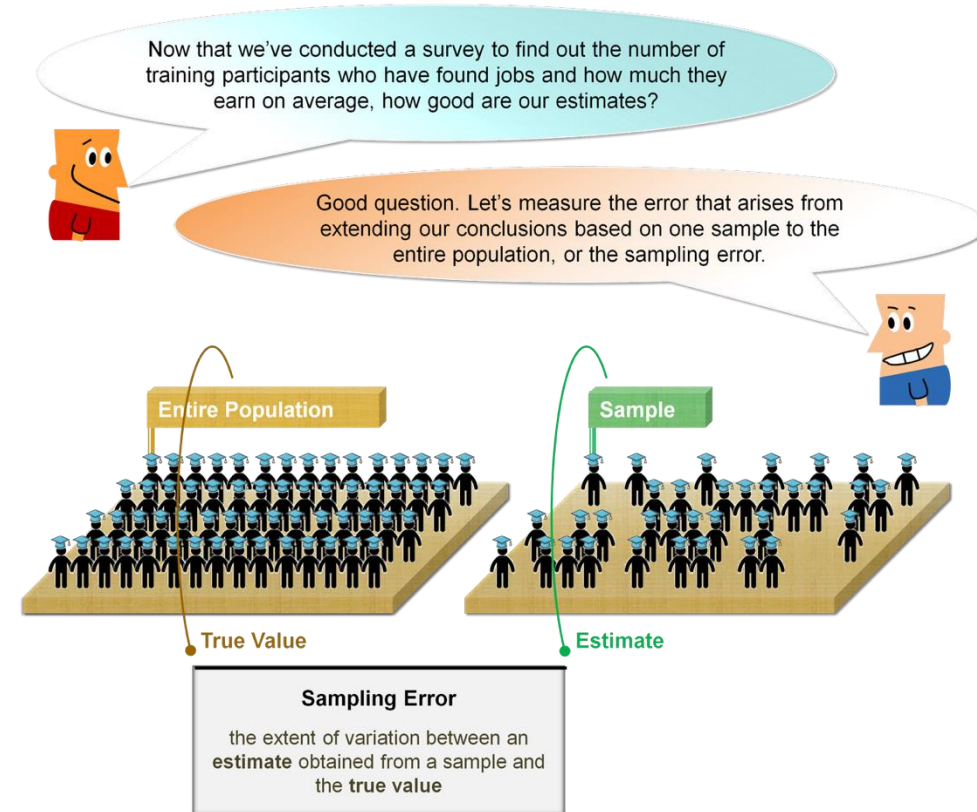
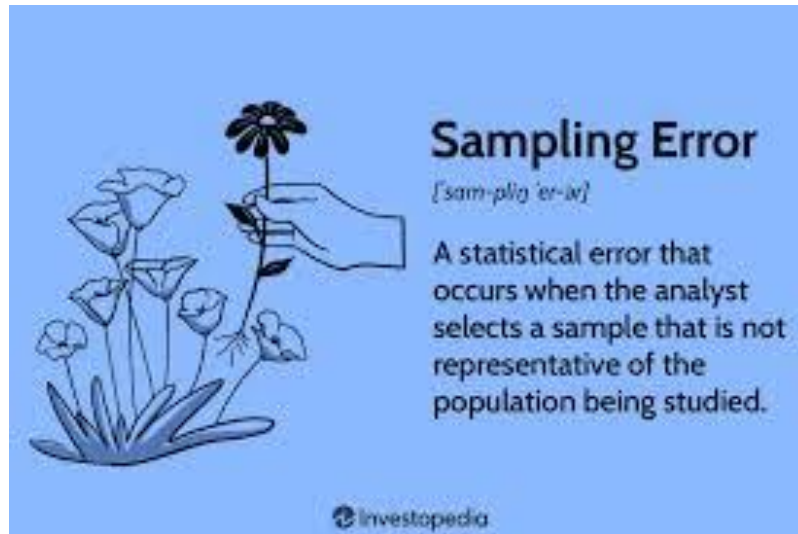
- a. Cluster
- b. Systematic
- c. Simple random
- d. Stratified

# Sampling Procedures





# Sampling Error





# Question

What is the scale type of the following survey question?

*Q: How satisfied are you with the overall food quality at Lulu's Malaysian Hawker in Melbourne?* A: 1. Extremely Dissatisfied; 2. Very Dissatisfied; 3. Somewhat Dissatisfied; 4. Neutral; 5. Somewhat Satisfied; 6. Very Satisfied; 7. Extremely Satisfied

- a. Ordinal
- b. Ratio
- c. Interval
- d. Nominal

# Measurement scaling

|                | Label | Order | Differences | Origin is 0 |                                       |
|----------------|-------|-------|-------------|-------------|---------------------------------------|
| Nominal scale  | ✓     |       |             |             | Non-metric /<br>Categorical<br>scales |
| Ordinal scale  | ✓     | ✓     |             |             |                                       |
| Interval scale | ✓     | ✓     | ✓           |             | Metric /<br>Quantitative<br>scales    |
| Ratio scale    | ✓     | ✓     | ✓           | ✓           |                                       |



# Nominal scales

Scales that only label objects; do not provide information about ordering

## *Examples*

Q. What is your occupation?

- 1) Digital marketer
- 2) Sales person
- 3) SEO specialist

Q. Have you visited Facebook in the past week?

- 1) Yes
- 2) No



# Ordinal scales

- Allow researchers to rank-order subjects' answers
- Enable us to know if the value of that variable increases or decreases
- We do not know if the differences in the order are equally spaced

## *Examples*

Q. How would you describe your usage of Facebook?

1. Non-user
2. Infrequent user
3. Regular user

Q. Please rank the following social media platforms based on your preference (1 = most favored, 3 = least favored)

1. Facebook
2. TikTok
3. Instagram

Q. What is your highest level of education?

1. No formal education
2. Primary school
3. Secondary school
4. Vocational education (TAFE)
5. University bachelor's degree
6. University postgraduate degree (master's, PhD)

Q. What is your approximate annual income? (before tax)

1. Less than \$20,000
2. \$20,000 - \$40,000
3. \$40,001 - \$60,000
4. \$60,001 - \$80,000
5. \$80,001 - \$100,000
6. More than \$100,000
7. Prefer not to say



# Interval scales

- Distance between descriptors is known
- Each value on the scale is assumed to be an ***equal distance*** apart
- We can interpret the magnitude of the differences in values directly
- The location of the zero point is not fixed, but arbitrary
- Place descriptors on the questionnaire evenly spaced apart to indicate interval

## Examples

Q. How satisfied are you with Facebook's customer service?

1. Extremely Dissatisfied
2. Very Dissatisfied
3. Moderately Dissatisfied
4. Neither Satisfied nor Dissatisfied
5. Moderately Satisfied
6. Very Satisfied
7. Extremely Satisfied

To what extent do you agree with the following statements:

|                                                                                | Strongly disagree     | Disagree              | Somewhat disagree     | Neither agree nor disagree | Somewhat agree        | Agree                 | Strongly agree        |
|--------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|-----------------------|
| Generally speaking, the higher the price of a product, the higher the quality. | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| The old saying "you get what you pay for" is generally true.                   | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| The price of a product is a good indicator of its quality.                     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| You always have to pay a bit more for the best.                                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |



# Ratio scales

There is an absolute “0” in the scale that is meaningful (a value of 0 means that particular variable is not present)

Provide the most information

## *Examples*

Q. How many friends/followers do you have? \_\_\_\_

Q. How many pages have you followed? \_\_\_\_

Q. Please indicate your annual income (before tax) \$\_\_\_\_

Q. Please indicate your age \_\_\_\_ years old

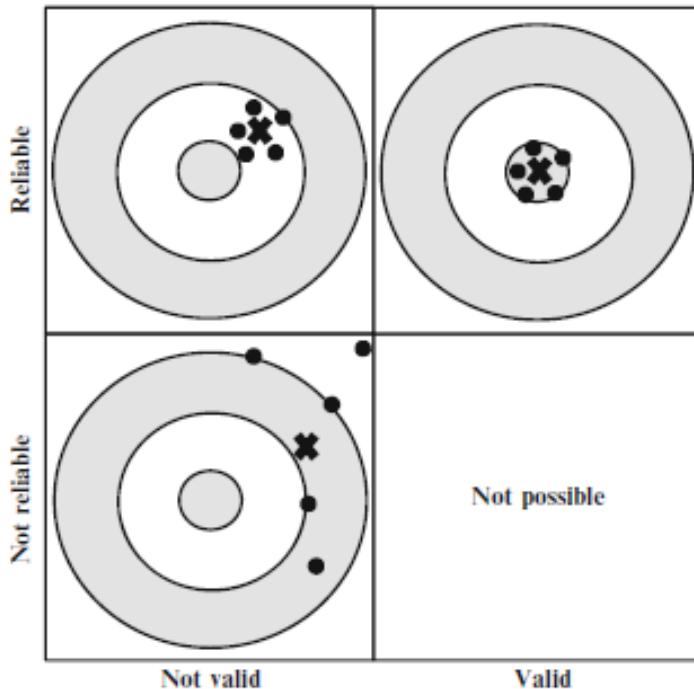


# Describe data

| Info level / adv. methods<br>↓ | Scale Level | Measure of Central Tendency | Measure of Variability | Type       |
|--------------------------------|-------------|-----------------------------|------------------------|------------|
|                                | Nominal     | Mode                        | (Frequency)            | non-metric |
|                                | Ordinal     | Median                      | Range                  |            |
|                                | Interval    | Mean                        | Standard Deviation     | Metric     |
|                                | Ratio       |                             |                        |            |



# Multiple item scales - reliability



True score model:  $X_O = X_T + E_S + E_R$

–  $X_O$  = the observed score

–  $X_T$  = the true score

–  $E_S$  = systematic error

–  $E_R$  = random error

Reliability

The degree to which what we measure is free from random error ( $E_R=0$ )

Validity

Whether we are measuring what we want to measure ( $E_S=0$ )

Generalizability

The extent to which one can generalize from the observations at hand to a universe



# Clean data - outliers

**Outliers are values situated far from all the other observations. There are three types:**

Outliers due to data collection or entry errors

Exceptionally high or low values that are a part of reality

Combinations of values that are exceptionally rare



**Should we delete or retain outliers?**

If they are data collection or entry errors: yes (or try to look up the correct values)

If they are not, retain and describe the impact of these outliers.

To detect outliers in the data, use boxplots or calculate z-values (  $z_x = \frac{X_i - \bar{X}}{S_x}$  ).

Rule of thumb:  $z < -3$  or  $z > 3$  are likely to be outliers.

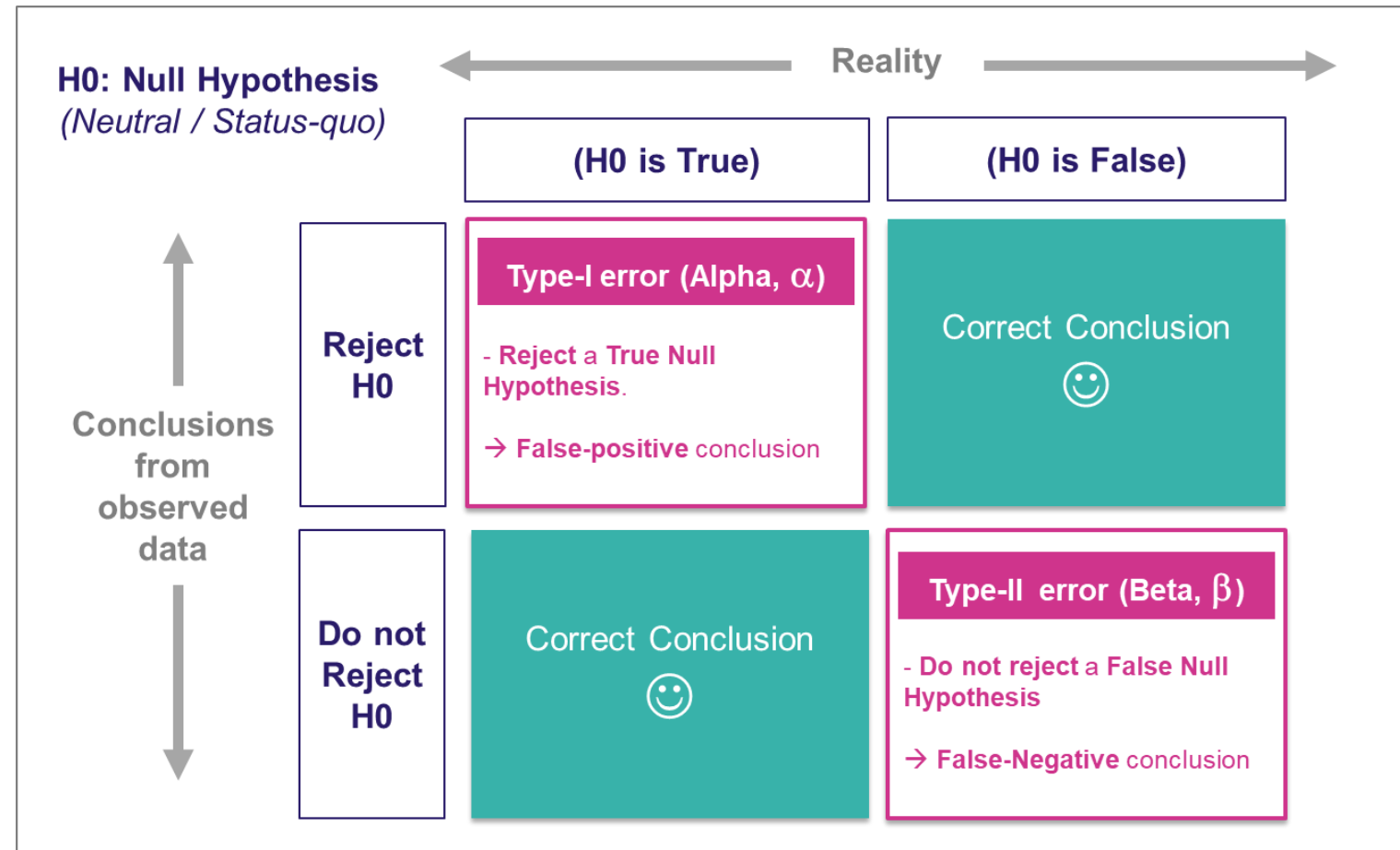


## Question

Which of the following is the correct *null hypothesis* ( $H_0$ ) for testing whether the average customer satisfaction rating is **4**? (Assume  $\mu$  represents the population mean of customer satisfaction.)

- a.  $H_0: \mu > 4$
- b.  $H_0: \mu < 4$
- c.  $H_0: \mu = 4$
- d.  $H_0: \mu \neq 4$

# Principles of Hypothesis Testing





## Question

A marketing research team wants to compare **customer satisfaction** with three sportswear brands: **Nike**, **Adidas**, and **Under Armour**. They surveyed **90 customers** (30 for each brand). The **average satisfaction scores** (on a 7-point scale) are:

Nike: **5.6** ( $n = 30$ ); Adidas: **5.2** ( $n = 30$ ); Under Armour: **5.0** ( $n = 30$ ); The **one-way ANOVA** produces a **p-value = 0.01**. At a significance level of  $\alpha = 0.05$ , what should the researchers conclude?

- a. At least one brand's satisfaction score is different.
- b. All three brands have different satisfaction scores.
- c. All three brands have the same customer satisfaction.
- d. Nike has significantly higher satisfaction than both Adidas and Under Armour.

# Testing for Differences

| Type of Tests              | Purpose of Test                                                                                                  |
|----------------------------|------------------------------------------------------------------------------------------------------------------|
| One Sample T-Test          | Compare the mean of a question answered by <b>one group</b> of subjects <b>against a constant</b> on metric data |
| Paired Samples T-Test      | Compare two means of <b>two different questions</b> answered by the <b>same group</b> of subjects on metric data |
| Independent Samples T-Test | Compare the mean of a question answered by <b>two different groups</b> of subjects on metric data                |
| ANOVA                      | Compare the means of a question answered by <b>more than two different</b> groups on metric data                 |



## Question

When should a researcher choose **Spearman correlation** over **Pearson**?

- a. When both variables are binary
- b. When data is paired and interval scaled
- c. When the variables are both nominal
- d. When one or both variables are ordinal



# Testing for Associations

| Types of Test                                  | Purpose of Test                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Chi-Square Analysis ( $X^2$ )                  | Test for relationship between two <u>nominal</u> variables.                                                |
| Spearman Rank Order Correlation Coefficient    | Test for relationship between two variables that are not nominal and that at least one is <u>ordinal</u> . |
| Pearson Product Moment Correlation Coefficient | Test for relationships between two <u>interval and/or ratio</u> variables.                                 |





## Question

In a multiple regression, if the **p-value in the ANOVA table** is much smaller than 0.05, this indicates:

- a. At least one predictor is significantly related to Y
- b. All predictors are significant
- c. The errors are normally distributed
- d. The intercept is different from zero

# Types of Regression Analysis

| Types of Test                 | Purpose of Test                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bivariate Regression Analysis | Analyze the linear relationship between <u>an</u> independent variable and <u>an</u> <b><u>interval/ratio</u></b> dependent variable.                      |
| Multiple Regression Analysis  | Analyze the linear relationship between <b><u>multiple</u></b> independent variables and <b><u>an</u></b> <b><u>interval/ratio</u></b> dependent variable. |

Note: the interpretation of R-square, ANOVA table, coefficient table, and the use of dummy variables.



# Exam Structure

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# Exam Structure

- Exam weighting: the exam accounts for **50%** of your final grade
- **Duration of the exam:** The exam will have a duration of 2 hours, and an additional 15 minutes allocated for reading time.
- **Exam type:** On-campus digital exam (using Canvas)
- **Exam format:** LMS Quiz with lockdown browser. Ensure that you have downloaded the browser on your computer for the exam.
- **Your task:** The exam will consist of MCQs and short/long-answer type questions. You need to answer all questions.
- **Authorised materials:** You are permitted to bring the textbook, paper-based notes, and a non-programmable calculator without added text ability to the exam venue.



# Final exam

- ❑ Bring your own devices/borrow a laptop from the university
- ❑ Need to borrow a laptop? Refer to this link for more information: <https://students.unimelb.edu.au/your-course/manage-your-course/exams-assessments-and-results/exams/laptop-preparation-on-campus-exams#before-swot-vac>
- ❑ Fill out and submit this form: [https://q.surveys.unimelb.edu.au/jfe/form/SV\\_3O5xUugzxyJM1eK](https://q.surveys.unimelb.edu.au/jfe/form/SV_3O5xUugzxyJM1eK)

## Don't have a reliable laptop to sit your exam?

If your personal laptop does not meet the requirements, you can borrow a University laptop for your centrally managed digital exams by **filling out this form**. Please note you will not receive a confirmation email after you complete this form. You will need to fill out this form for each exam you require a laptop for.



# Exam structure

- The exam contains 7 MCQs + 6 short/long-answer questions.
- Some questions have sub-questions.
- Answer all questions.
- Be precise and succinct in your answers
- Bullet points are encouraged for answers



# Multiple Choice Questions

- Refer to quizzes



# Short and long answer questions

- You will be presented with short scenarios in each question
  - You will be asked to recommend a solution in a specific aspect of marketing research (*what we covered in class*)
  - You need to justify your solution
  - You need to describe the solution
  - You need to interpret the possible implications of your solution
  - You can make valid assumptions about the research context.
- 
- Example: Given the scenario ..., please recommend the most suitable sampling technique and provide your reasoning.





# Short and long answer questions with a mini case

Example:

- A research survey questionnaire, along with research objectives, may be provided.
- Based on the survey and research objectives, you may be asked multiple research design questions and statistical recommendations, including
  - ✓ **Developing appropriate hypotheses,**
  - ✓ **Recommending appropriate statistical tests,**
  - ✓ **Choosing appropriate survey questions for research objectives**
  - ✓ **Interpreting data results**
  - ✓ **Developing survey items**
  - ✓ **Etc.**



# Sample questions

Review a survey design and answer the following questions.

- Q1: “how satisfied are you with the quality of product A?”  
*(Measured on 5-point scale; 1-very dissatisfied; 5-very satisfied)*
- Q2: “how satisfied are you with the style of product A?”  
*(Measured on 5-point scale; 1-very dissatisfied; 5-very satisfied)*
- Q3: “what is your overall satisfaction with product A?”  
*(Measured on 5-point scale; 1-very dissatisfied; 5-very satisfied)*
- Q4: “what do you usually use product A for?”  
(\_\_ business; \_\_ leisure; \_\_ both business and leisure)
- Q5: “would you recommend product A to friends?” (\_\_ yes; \_\_ no)
- Q6: “what is your gender?” (\_\_ male; \_\_ female)



# Sample questions

1. If the researcher wants to test whether consumers are more satisfied with the quality of product A than with its style, which statistical test would you perform and why?
2. If the researcher wants to test which of the two factors: quality satisfaction and style satisfaction, has a greater impact on consumers' overall satisfaction with product A, which test would you perform and why?



## Some exam tips

Provide a clear rationale and reasons behind your response.

- Justify your response for good marks

If you need to justify your recommendations/responses 'X', consider the following structure:

- List reason(s) for selecting 'x'
- Why is 'x' better than the alternative option 'y'?
- To strengthen the argument, consider providing examples (if possible)



# Exam Tips

- Think before you write
- Allocate your time wisely
- All questions are intended to be self-explanatory, and minimal clarification will be provided during the exam. If you feel it necessary, explicitly write down your assumptions.
- Get the point/logic through instead of beating around the bush
- Do not provide vague or generalized responses
- Submit the answers before the time is up



# Student Learning Surveys (SLS)

Results from the Student Learning Surveys (SLS) provide valuable feedback to the University and to your Subject Coordinators for subject improvements for future cohorts!

Log in directly from your SLS notification email or visit [unimelb.edu.au/sls](https://unimelb.edu.au/sls).

All end-of-semester surveys are:

- Completely anonymous – all entries are de-identified
- Online and available for three weeks
- Available to be completed on computer or your mobile

More information can be found in the student:

FAQs: [unimelb.edu.au/sls/students/faqs](https://unimelb.edu.au/sls/students/faqs)

**Monday 19 May until Sunday 8 June 2025.**

Please take ~10 minutes to complete the survey now! 🙏



## Some take away thoughts

- Think about the importance of marketing research in today's world
- Think about planning, designing, and conducting a market research study
- Think about the skills you have developed by studying this subject
- Including analytical skills, statistical skills, locating and synthesising information, working with a team, and communication
- Think about what you have learned from working on a research project and working with a group of people